Math and Theory: collection overview

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Abstract

The catalogue contains thirteen result families and thirteen preprints in mathematics, theoretical computer science, and mathematical physics. Twelve papers have complete Lean proofs within their stated formal scopes. The thermal proof covers finite spectral statements, without a formal stochastic long-time identification theorem. The Gyires preprint proves inequalities for structured matrices and develops related polynomial bounds. Its principal order-five and order-six claims, the all-order permutation-polynomial and four-block theorems, and the exceptional-vector theorem have complete Lean proofs. The results comprise counterexamples, exact special cases, and sharp sequence bounds. The collection makes no claim of first priority. The papers have not been peer reviewed.

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1 How to use the collection

The manuscript catalogue links each paper to its formal statements and independent computations. The verification guide records the commands and their results. Seven families share a Lake project pinned to Lean 4.35.0. Its 29 selected Comparator statements include equality, sharpness, and auxiliary positivity claims. The decision-tree and thermal packages use separate Lean 4.19.0 environments and a pinned Mathlib revision. The decision-tree package has 29 axiom reports of its own. This is a separate count from the shared Comparator selection.

2 Mathematics

2.1 Real positive-definite counterexamples to extended q-permanent monotonicity

For every real parameter $q > 1$, some real symmetric positive-definite matrix has a negative q-permanent derivative at q. An explicit matrix of order five decreases from $q = 14$ to $q = 15$. The construction also gives correlation matrices with no zero entries and integer-entry examples. The dimension can depend on q. The real-symmetric question on $[-1, 1]$ remains outside the result.

Scope: Counterexample and all-parameter extension.

Manuscript Formal statement and prior work.

2.2 Gyires's permanent inequality for structured matrices

Every nonnegative real circulant matrix A of order five satisfies $24\,\mathrm{per}(AA^T) \le 625\,\mathrm{per}(A)^2$. Equality holds exactly for constant matrices. This proves the order-five circulant case of Gyires's ratio inequality. A bound in sorted-gap coordinates has sharp coefficient 484. The conjecture for arbitrary doubly stochastic matrices remains open.

Scope: Structured matrix inequalities and related polynomial bounds. The principal order-five and order-six claims, the all-order permutation-polynomial and four-block theorems, and the exceptional-vector theorem have Lean proofs.

Manuscript Formal statement and prior work.

Broader arguments in the same preprint

The manuscript also proves results for circulants of orders six and seven, with sharp sorted-gap margins. It treats $aI + bQ + cQ^2$ in every dimension, where Q is any permutation matrix and the coefficients are nonnegative. Further results cover four-block doubly stochastic matrices with unequal block sizes and boundary entries, matrices with one exceptional column or row, and every rank-at-most-two doubly stochastic matrix through order fifteen.

The supporting sections prove structural, scalar, deletion, balancing, local polynomial, and configuration-model results. Exact counterexamples identify the limits of auxiliary assertions. The complete support record gives the source map and exact reproduction commands.

The all-order four-block and exceptional-vector theorems have complete Lean proofs. Their inequalities and equality cases are among the 29 selected formal statements. The principal order-seven and finite rank-two theorems remain incomplete in Lean. Separate Lean packages prove the principal order-six and all-order permutation-polynomial theorems. The all-order rank-two problem and the unrestricted Gyires conjecture remain unproved here.

2.3 An entropy increase for a structurally disassortative DomiRank graph

A connected graph with 14 vertices and 32 edges has negative degree–log-degree edge covariance. All resolvent scores are positive on $[251/1000, 63/250]$ inside the stable domain. Its entropy nevertheless increases by more than $1/10000$ in natural-logarithm units. This refutes the source's assertion for every positive-score interval.

Scope: Counterexample to an interval monotonicity conjecture.

Manuscript Formal statement and prior work.

2.4 Diagonal concavity does not imply concavity for sums of elementary symmetric polynomials

For $M > 0$, let $f_M = 1 + M\sigma_1 + \sigma_2 + M\sigma_3 + \sigma_4$ in four variables, where σ_j is the elementary symmetric polynomial of degree j. The fourth root $f_M^{1/4}$ is concave on the positive diagonal exactly when $M \geq 1$. It is concave on the full positive orthant exactly when $M = 1$. Thus every $M > 1$ disproves Lachaume's criterion with strictly positive coefficients.

Scope: Counterexample and complete parameter classification.

Manuscript Formal statement and prior work.

2.5 Sharp root log-concavity of alternating Motzkin sums

Let $T_n = \sum_{j=0}^{n} (-1)^{n-j} M_j$, where M_j is the Motzkin number. The sequence $T_n^{1/n}$ is strictly log-concave on its tail starting at five. The inequality reverses at center five, so no earlier tail is log-concave. This strengthens Zhao's proposed tail starting at nine. The source recurrence and finite sum are linked in Lean.

Scope: Conjecture resolution with an optimal threshold.

Manuscript Formal statement and prior work.

2.6 Counterexamples to a late-iterate ordering conjecture for generalized Hofstadter functions

For every $k \geq 4$, some input n satisfies $F_k^{\circ(2k+1)}(n) > F_{k+1}^{\circ(2k+2)}(n)$. Here the superscript denotes repeated composition. The proof constructs inverse iterates with exact gap $k-3$. The witness $F_4^{\circ 9}(132) = 8 > 7 = F_5^{\circ 10}(132)$ gives the first instance. The result concerns infinitely many parameters. It does not assert infinitely many violating inputs for each fixed parameter.

Scope: Counterexamples for every parameter k≥4.

Manuscript Formal statement and prior work.

3 Theoretical computer science

3.1 Counterexamples to a best-of-n win-rate bound at every budget

For every $n \geq 2$ and $\delta > 0$, a finite positive reference distribution with largest atom at most δ violates the proposed best-of-n divergence–win-rate curve. A four-response best-of-two example gives exact probabilities and a strict logarithmic certificate. The all-budget proof uses finite interval bounds and gives a quantitative positive gap. It does not determine the sharp replacement curve.

Scope: Disproof for every budget n≥2 and every positive atom cap.

Manuscript Formal statement and prior work.

3.2 Logarithmic influence bounds and variance counterexamples for Boolean decision-tree covariance

Let T be a finite Boolean decision tree under uniform independent bits. No coordinate repeats on a root-to-leaf path. Let $k_i(T)$ count all query nodes labelled i, and let $I_i(f)$ be the probability that flipping coordinate i changes the output. The weighted sum of absolute child covariances satisfies

$$\sum_v 2^{-d(v)} \left|\mathrm{Cov}(f_{v,+}, f_{v,-})\right| \le \sum_i I_i(f) \log_2 \max\{1, k_i(T)\}.$$

Here $d(v)$ is the node depth, and the two child functions use the same input cube. The read number counts nodes in the entire tree, not queries along one path.

For a read-k tree, the bound is $I(f) \log_2 k$. The dummy-prefix construction of Wan, Wright, and Wu attains equality when k is a power of two. The logarithmic potential is due to Shalev. A four-variable counterexample has covariance 5/4 and variance 3/4. A nine-variable counterexample has covariance 4863/4096 and variance 3007/4096, with every query relevant. Both refute the coefficient-one logarithmic variance proposal for a given tree.

Scope: Covariance inequalities, an equality family, and exact finite witnesses. No Fourier-entropy theorem is formalized.

Manuscript Prior work Formal proof and reproduction.

4 Mathematical physics

4.1 A three-state thermal counterexample to a heat-current kurtosis conjecture

A three-state rate model with two positive-temperature baths satisfies local detailed balance. Every total off-diagonal rate is positive. The tilted generator records heat transferred to the cold bath. Its dominant eigenvalue branch is four times continuously differentiable near zero and has a larger real part than every distinct complex eigenvalue. The exact derivative ratio is

$$\frac{C_4}{C_2} = -\frac{3470947982622836}{176627429595846691} < 0,$$

where C_n is the nth derivative at zero. This refutes Eq. (19) of Ptaszyński, *Physical Review E* **106**, 024119 (2022), in the paper's finite thermal spectral formulation. The third derivative is positive, so the witness does not refute the separate skewness conjecture.

Scope: Finite rate matrices, local detailed balance, complex eigenvalues, and derivatives of a local dominant branch. The stochastic long-time theorem that identifies the branch with a cumulant generating function is not formalized. No minimal-state or empirical validation claim is made.

Manuscript Prior work Formal proof and reproduction.

5 Discrete and quantum bounds

5.1 The sharp Fourier bound at density one eighth

For every indicator of mean 1/8 on a finite uniform Boolean cube, the level-one Fourier weight is at most 3/64. A codimension-three subcube attains this bound in every dimension at least three. The proof also gives the equivalent sharp average Hamming-distance bound. It uses exact rational polynomial certificates and a dimension-independent moment identity.
Manuscript Formal proof and reproduction.

5.2 Domination sequences of powers of paths and cycles

The domination counts of every natural power of a path or cycle form an ultra-log-concave sequence. Dividing each count by its binomial coefficient gives the normalized log-concavity inequality. The proof includes all natural graph orders and powers. It does not assert the same property for arbitrary graphs or real-rootedness of their domination polynomials.
Manuscript Formal proof and reproduction.

5.3 Universal entanglement embezzlement

A target-independent catalyst of local dimension 2^{4L} gives root infidelity at most $125m/L^2$ for every Schmidt probability vector on m entries and $L \geq 1$. Zero entries are permitted. The proof constructs local basis permutations and gives their finite complex Hilbert-space interpretation. It does not separately formalize arbitrary changes to an external Schmidt basis, gate efficiency, or catalyst reuse.
Manuscript Formal proof and reproduction.

5.4 The asymmetric Merino–Welsh region

The theorem characterizes every nonnegative pair (x, y) for which

$$T_M(x, 0)T_M(0, y) \geq T_M(1, 1)^2$$

holds for all finite matroids without loops or coloops. The criterion uses two explicit scalar inequalities for every integer $k \geq 2$. The proof includes empty ground sets and finite ground sets in arbitrary ambient types. The package fetches Liu's pinned proof library from its public repository.
Manuscript Formal proof and reproduction.

6 Verification and authorship

Lean checks the proofs against their formal statements. In the shared Lean 4.35.0 library, separate Challenge modules state the intended claims. Comparator checks these against their proved Solutions. The verification record identifies the completed runs of Lean, NanoDa, and con-ron for those seven families. The six standalone study packages have their own pinned Lean environments and source compilation records. Their audit logs state the permitted axioms. Exact arithmetic programs check the accompanying finite certificates. The shared Comparator results do not cover those packages. Independent exact and numerical programs test finite certificates and examples. These finite computations do not replace the general proofs.

This work is fully autonomous agentic research generated by AI systems. AI agents produced the mathematical arguments, formal proofs, computations, and manuscripts. Dev D. Goyal is responsible for publication. The authorship disclosure identifies these tools and their roles. These preprints have not been peer reviewed.